



Fig. 3: Primary haptic design criteria

‘reverse-electro vibration.’

You may have experienced a very basic version of this technique if you use a unibody aluminium Macbook Pro notebook computer. Dragging a dry finger over the laptop creates a characteristic rubbery feeling, although you are actually touching aluminium. This is caused due to the capacitive set-up created by the finger and the metal surface that attracts the finger.

You can create this effect on a system by giving weak current to the device being used or, more specifically, to the area of the device that is being used for interaction. If the material is conducting, a capacitive set-up will be created when a dry finger touches it. The capacitive set-up will generate an oscillating electric field around the skin and the finger tips, which then go on to create a variable sensation of friction depending on the frequency and amplitude of the applied signal. The ability to control friction by varying the frequency and amplitude will allow to create different ‘virtual’ surfaces as you desire. These surfaces are called virtual because they are only felt to be different—the actual surface remains the same (aluminium in the example above).

Design tips you need to keep in mind

The main challenge in designing a haptic device is to make the control interface feel exactly like the tool being used originally by the operator.

“Take the case of a haptic device that mimics the tools of an endo-

scopic surgeon. The haptic device has to feel like the tools of an endoscopic surgeon. The actuators and sensors have to be implemented to look almost like that. The major issue would be balancing the weight of different actuators along different axes so that the surgeon feels like he is using his original equipment,” explains Biju R. Varkey, CEO, Designs and Projects Development.

“Another design aspect would be to provide kinetic motion and freedom exactly like he would feel while conducting the said operation on an actual patient. It is also very important to make the software and control mechanism fast and accurate,” he adds.

Haptics applications can be extremely complex as they are expected to produce response to human touch. Therefore integrating haptics into any device requires a confluence of system-level design expertise.

“Appropriate actuators and their respective locations must be selected carefully as these are crucial from an end-user experience point of view. Additionally, software written for the device should not only control but also optimise the actuators to ensure that the quality of the haptics’ sensations is near to reality. Giving consistent experience to the user could be a success factor for such a device. It is important to recognise that designing haptics into any device requires a system-level approach,” advises Tamboli.

“The value of a well-designed multimodal interface can provide great differentiation and an enhanced user

experience to consumers. In short, haptics is all about giving feedback to the user in touch-sensation form, and hence it may have cultural influence too. For instance, softer touch-sense might mean one thing to one type of users while it would mean different (or lesser) to others. Therefore user study and adaptation of the device should be taken critically as it would define success or failure of the device with haptics,” he said.

Satwinder Singh, senior design engineer, Designs and Projects Development, explains, “Recent advances in haptics have added more degrees of freedom of motion to the haptic device with more and very precise control of force being exerted. These devices have very precise motion and force sensors to determine the actions being taken by the user.”

“Fig. 4 shows the design of a three degrees of freedom haptic device that we have developed. This device is capable of withstanding two rotary axes and one linear axis. The design is made so that the whole mechanical structure is balanced at the centre of the whole device. This is important because unbalanced weight can cause the user to feel the weight as force. If this was not balanced by design, we would have had to pre-force it in the other direction to compensate for the weight imbalance, thus causing more delay and more computation,” he adds.

What’s next?

One of the most welcomed applications of haptics technology is in medicine. Utilising haptics in medical

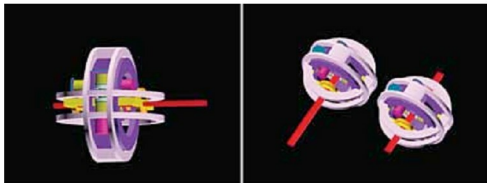


Fig. 4: 4-axis haptic concept by Designs & Projects Development